

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/G_DEB_SAAWYw")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 33113, C2 = 6666, C3 = 2212)

text_priors <- '
Distrib (Fm,          TruncNormal,          0.1,      0.2, 0.01,      0.8);
Distrib (pAm,         TruncNormal_cv,        100,      0.2, 40,      2000);
Distrib (r_pAm_pM,    TruncNormal_cv,        0.8,      0.1, 0.1,      2);
Distrib (v,           TruncNormal_cv,        0.18505, 0.1, 0.001, 2);
Distrib (kap,         Uniform,                0,        1);
Distrib (kapA,        Uniform,                0.1,      0.9);
Distrib (Ehp,         TruncNormal_cv,        100,      0.2, 40,      300);
Distrib (E_coc,       TruncNormal_cv,        30,      0.2, 1,      300);
Distrib (rOM_ClxHorse, LogUniform,          0.00001, 0.3);
Distrib (dv,          TruncNormal_cv,        2,      0.5, 0,      10);
Distrib (w,           LogUniform,            0.1,      100);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	Fm.1.	pAm.1.	r_pAm_pM.1.	v.1.	kap.1.	kapA.1.
Result_mode	0.48955900	163.70000	0.78960800	0.18617500	0.91814400	0.70923600

2.5%	0.35170290	138.18400	0.59853200	0.15166200	0.87314965	0.58178115
97.5%	0.70690340	187.26700	0.87478710	0.22448600	0.94451500	0.81169800
Min	0.24323600	90.58550	0.50345200	0.12413900	0.43482600	0.19504900
Max	0.79798000	210.00700	0.96360900	0.25243500	0.96420800	0.86845800
Result_mean	0.52140633	160.06789	0.74722390	0.18783224	0.91296040	0.68501359
Result_sd	0.09129444	12.64168	0.06714888	0.01872845	0.01814403	0.05638819
	Ehp.1.	E_coc.1.	rOM_ClxHorse.1.	dv.1.	w.1.	Sigma_W.1.
Result_mode	102.64400	16.539600	0.003793150	2.6644400	1.3134500	0.11524700
2.5%	69.01114	10.228600	0.001898890	1.7401400	0.1480050	0.10455900
97.5%	144.08200	26.348800	0.005599490	3.8317200	2.6830645	0.13259880
Min	45.10530	7.673000	0.000125481	0.5220400	0.1000210	0.09538670
Max	182.54100	34.450600	0.187905000	4.5576700	3.6533400	4.53346000
Result_mean	104.19857	17.380041	0.003471068	2.6990222	1.3598440	0.11804364
Result_sd	18.20460	4.079784	0.001426577	0.5493936	0.6967378	0.03969013

Number of observations, n = 277
 Number of parameters, k = 11
 Log-likelihood, LL = 0.630816
 BIC = 60.60256

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	1.01	1.02
E_coc.1.	1.11	1.30
Ehp.1.	1.14	1.41
Fm.1.	1.09	1.26
kap.1.	1.07	1.22
kapA.1.	1.09	1.27
pAm.1.	1.15	1.41
r_pAm_pM.1.	1.03	1.04
rOM_ClxHorse.1.	1.05	1.12
Sigma_W.1.	1.00	1.01
v.1.	1.00	1.02
w.1.	1.07	1.23

Multivariate psrf

1.17

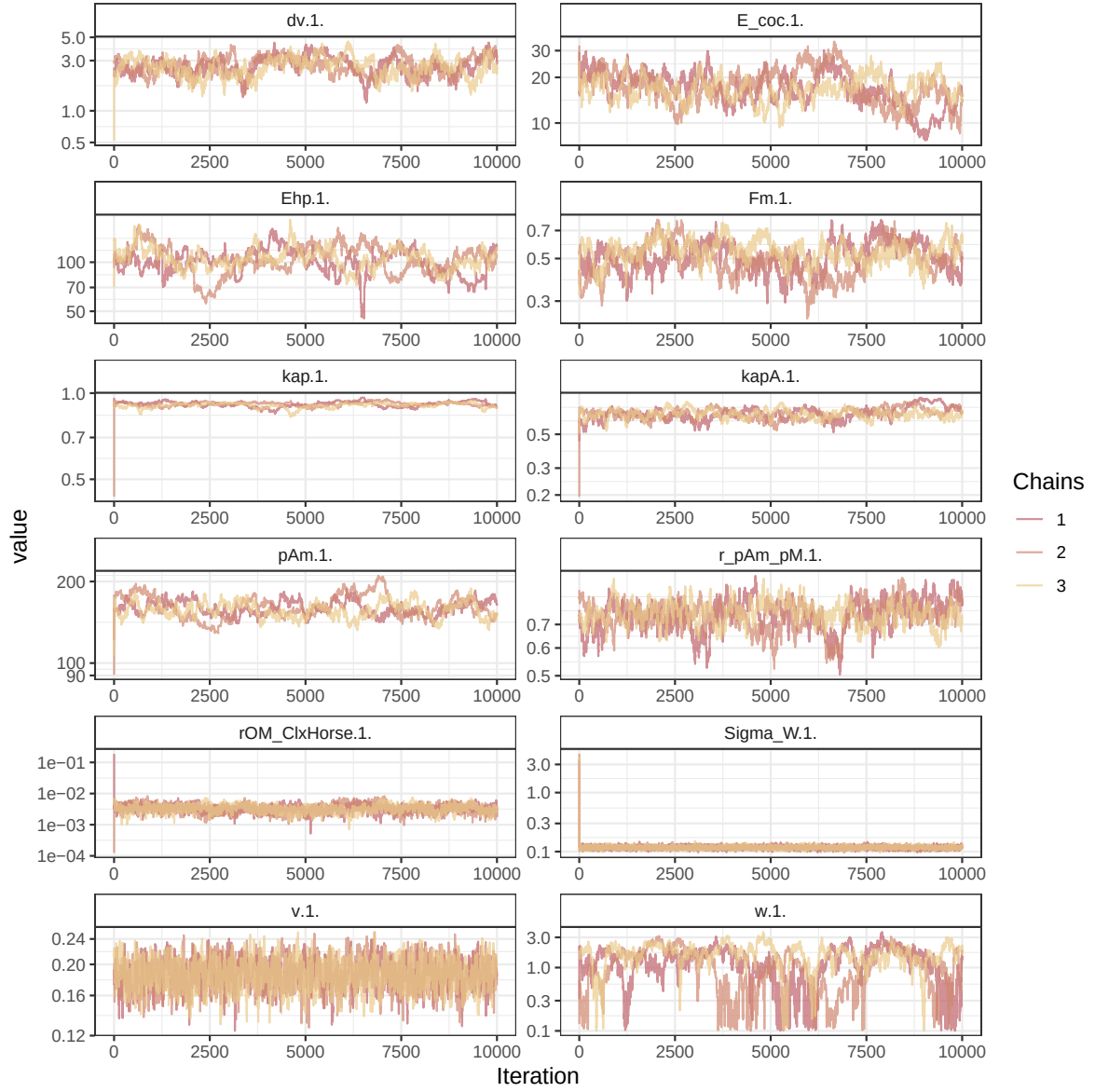


Figure 1: Parameters chains. Only the first iteration and the last r `Nb_iter_kept` are represented.

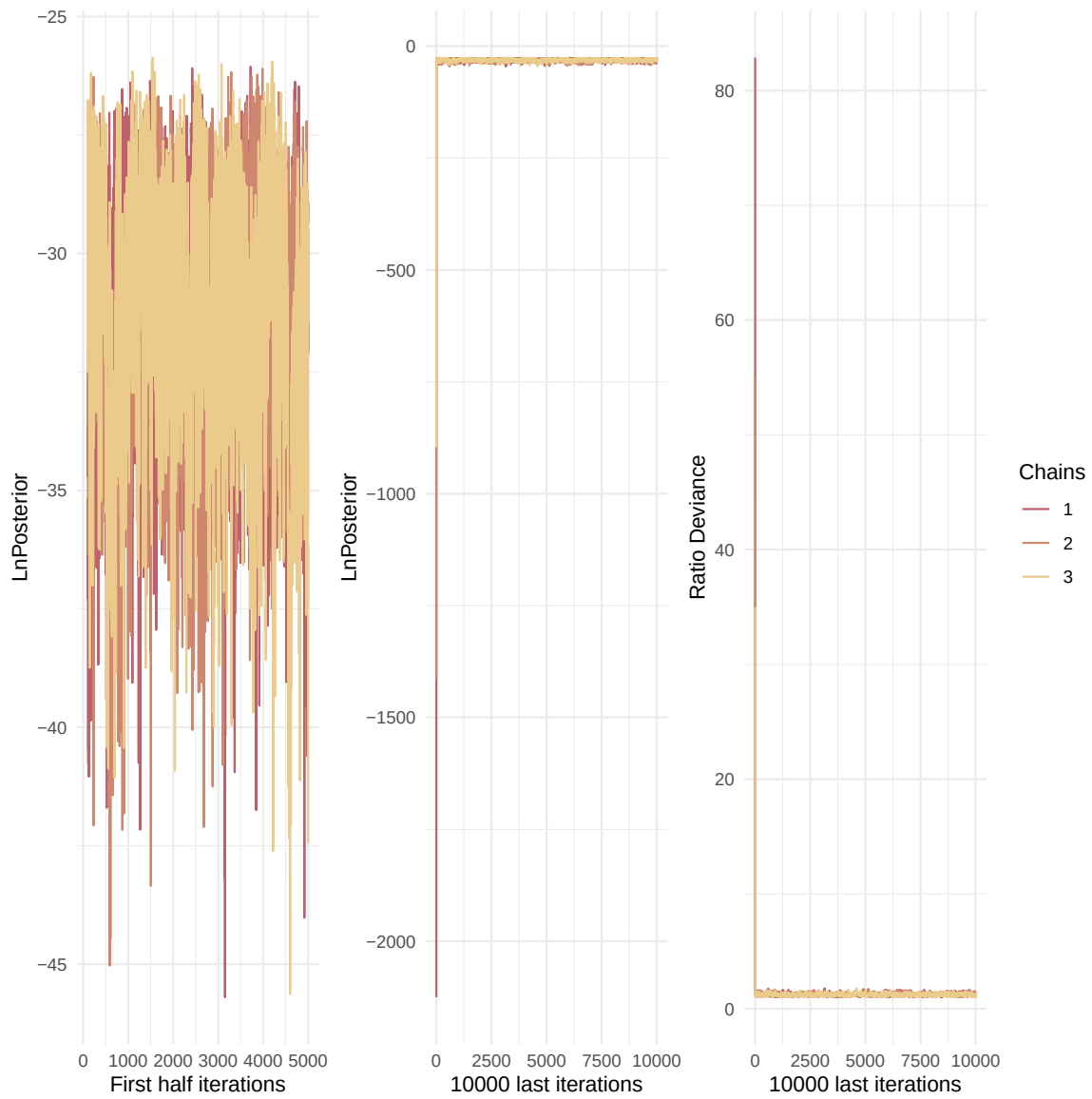


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

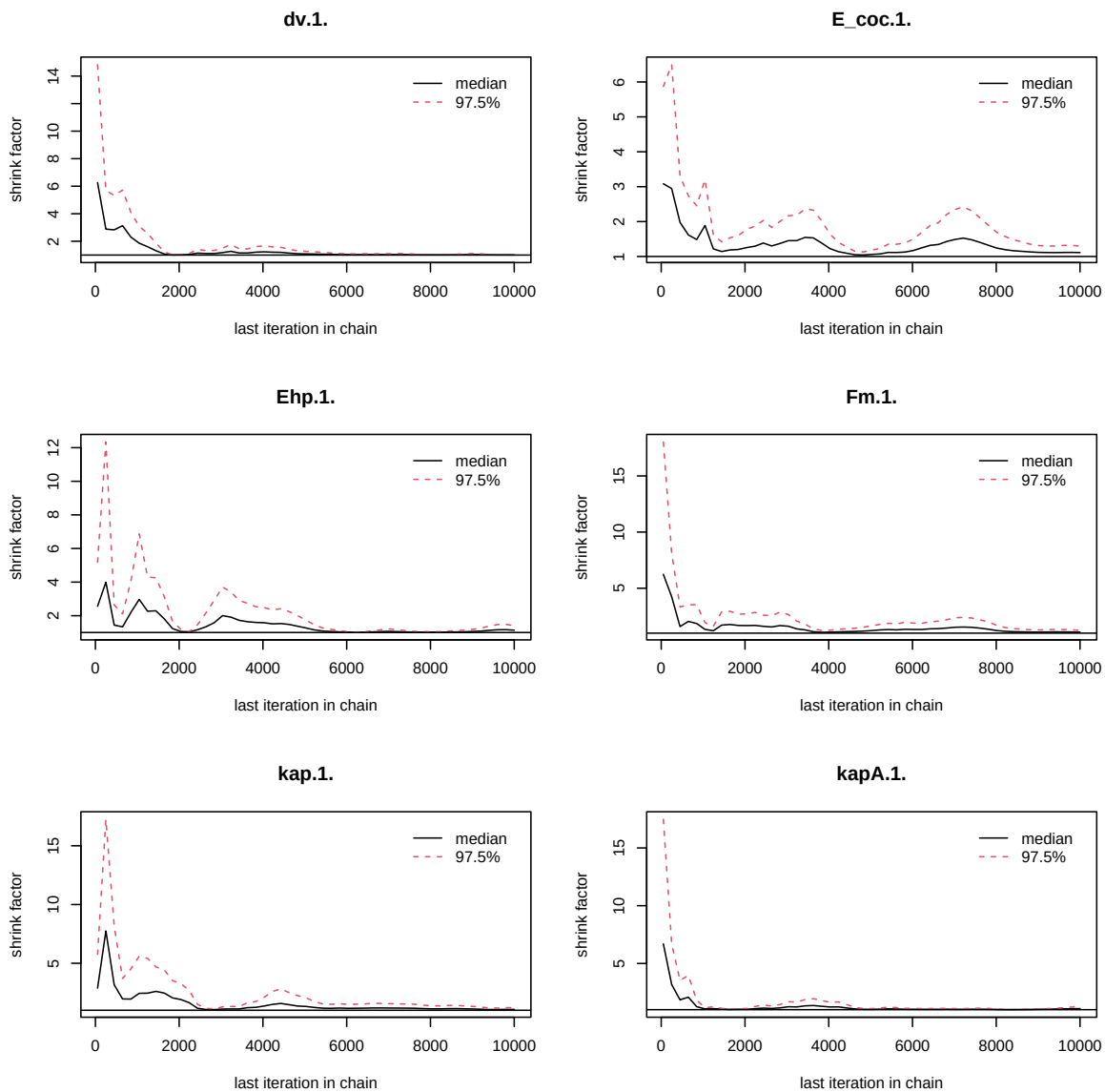


Figure 3: Chains convergence.

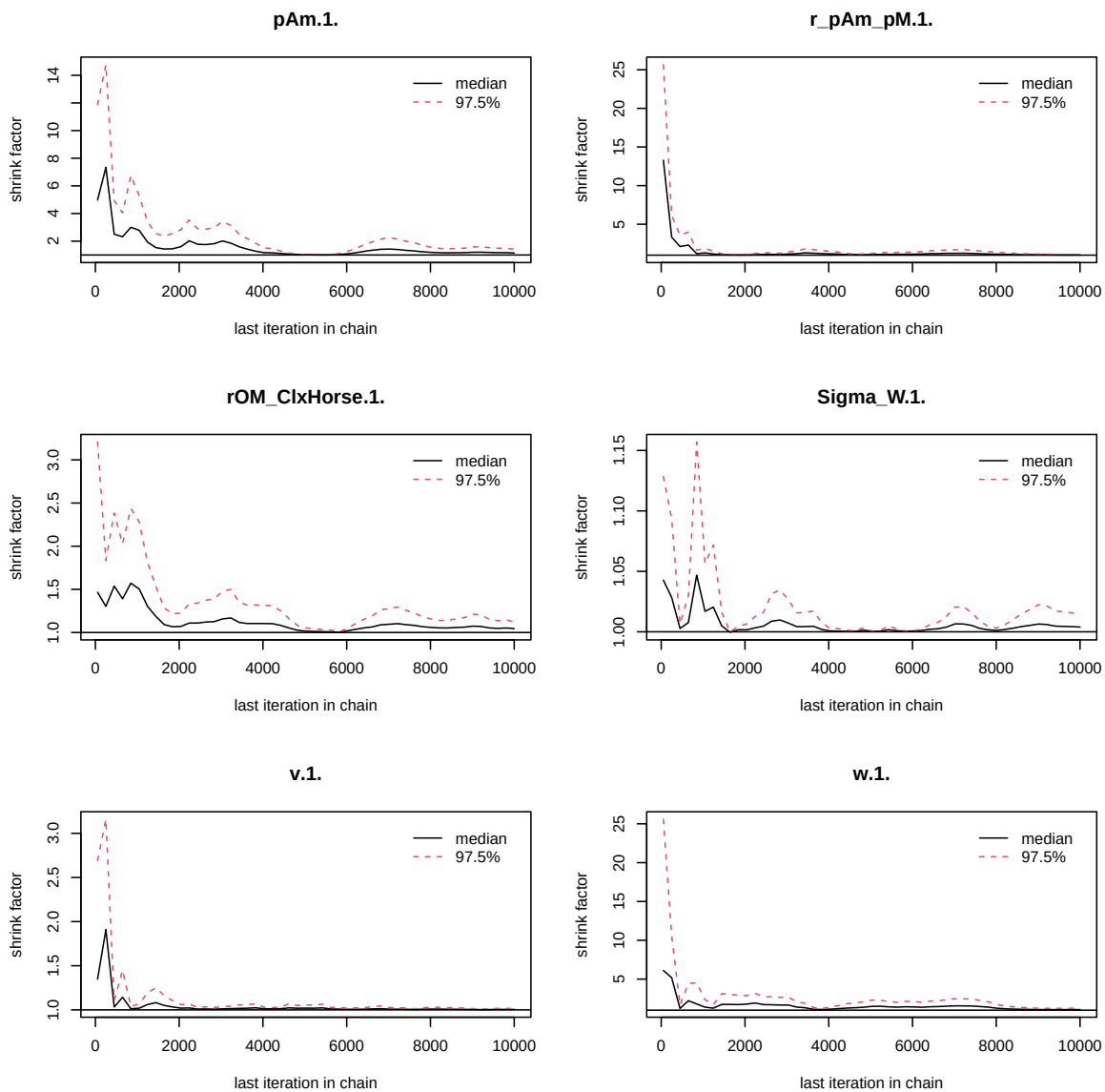


Figure 4: Chains convergence.

```

mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)

```

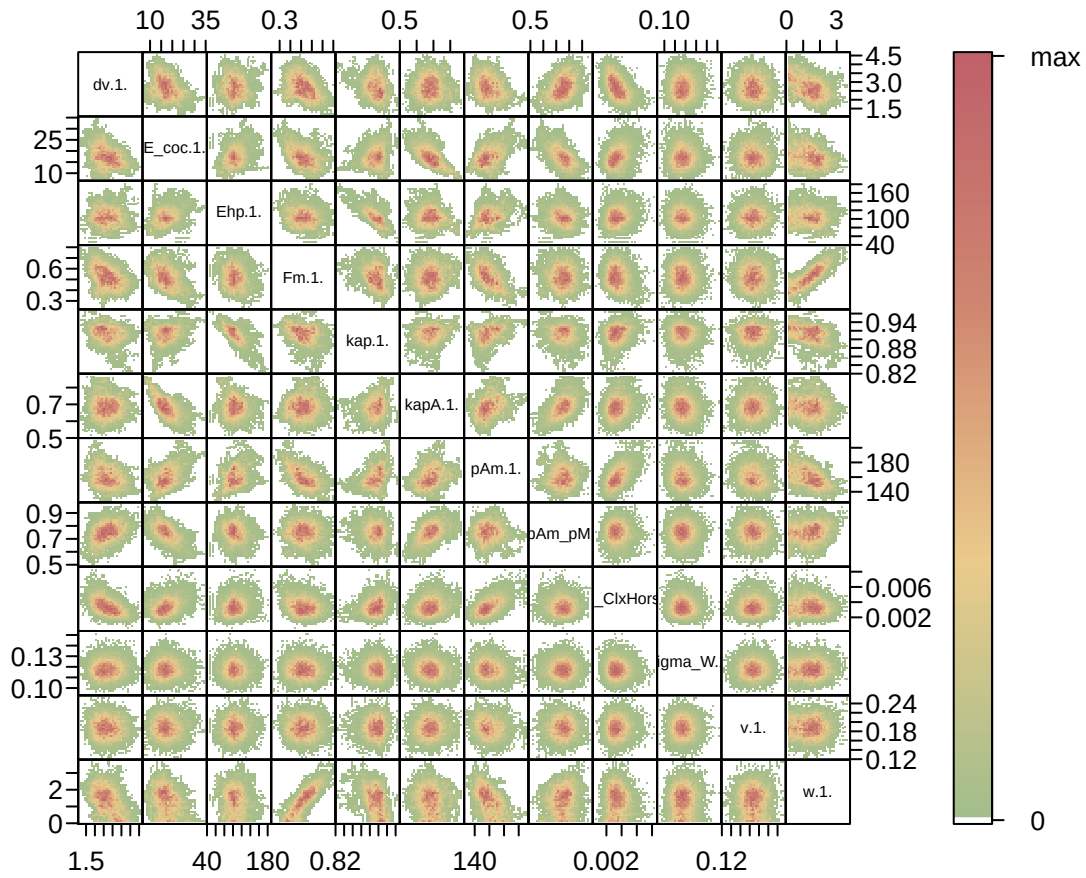


Figure 5: Parameter correlations.

**Posterior
and prior distributions**

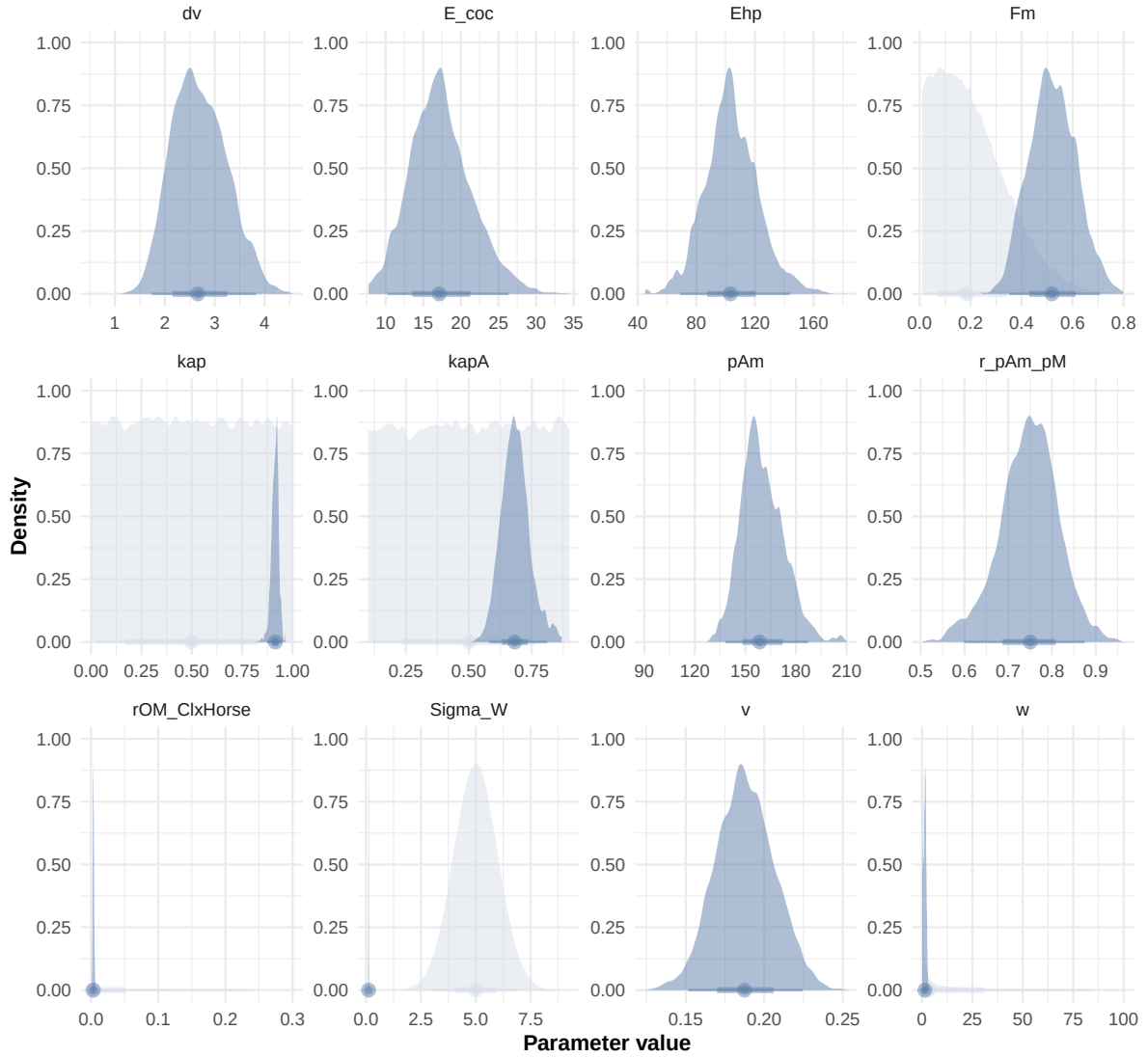


Figure 6: Distributions of the priors and the posteriors.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\\\.1\\\. $|\\.1\\\. ", "", param),
    line = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
Length_struct,
Reproduction,
Energy,
Organic_matter,"
```

```
f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_print)
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_print)

# ./mcsim.DEBcali DEB_sim_full.in
```

```
#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
)
```

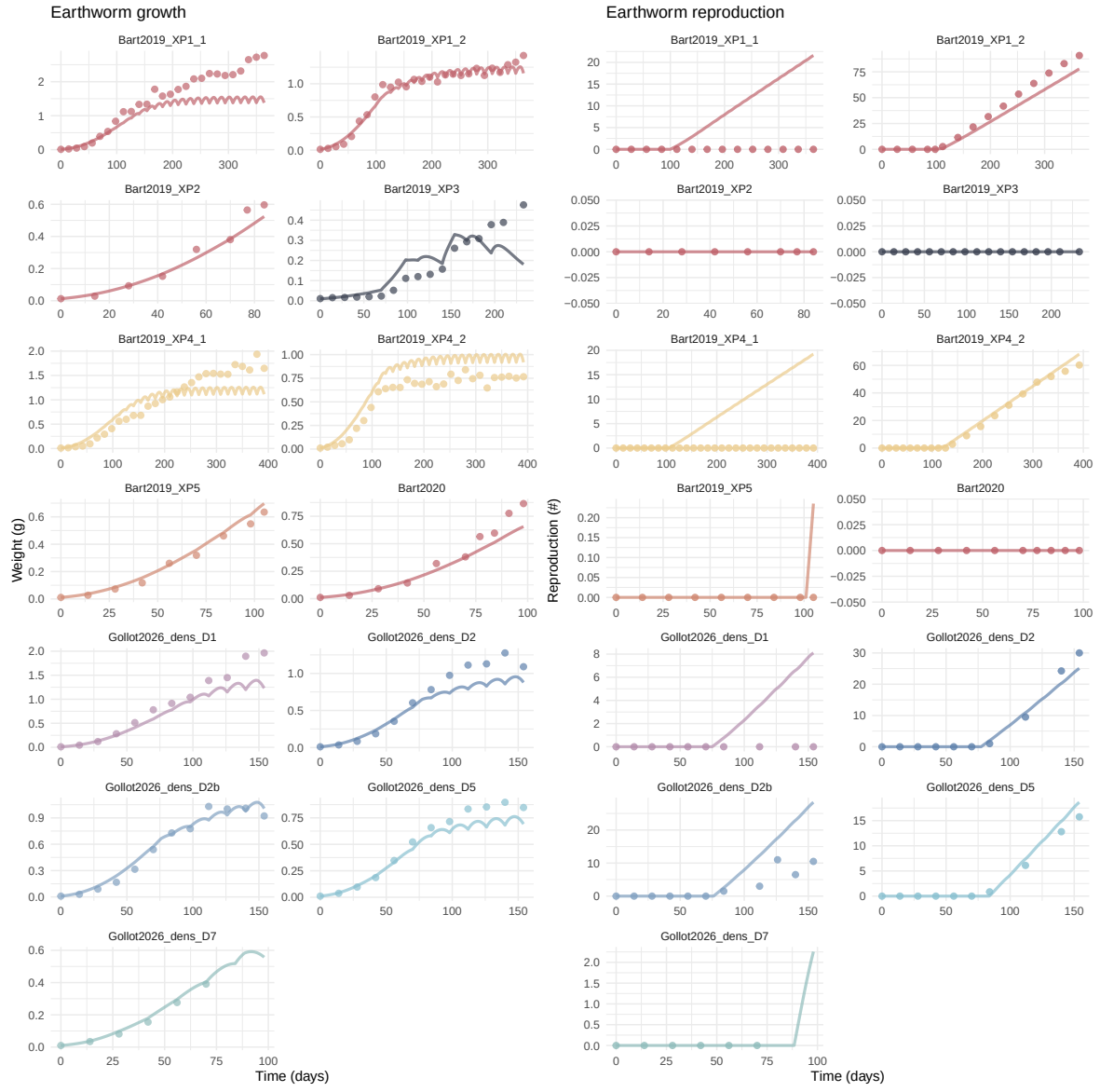
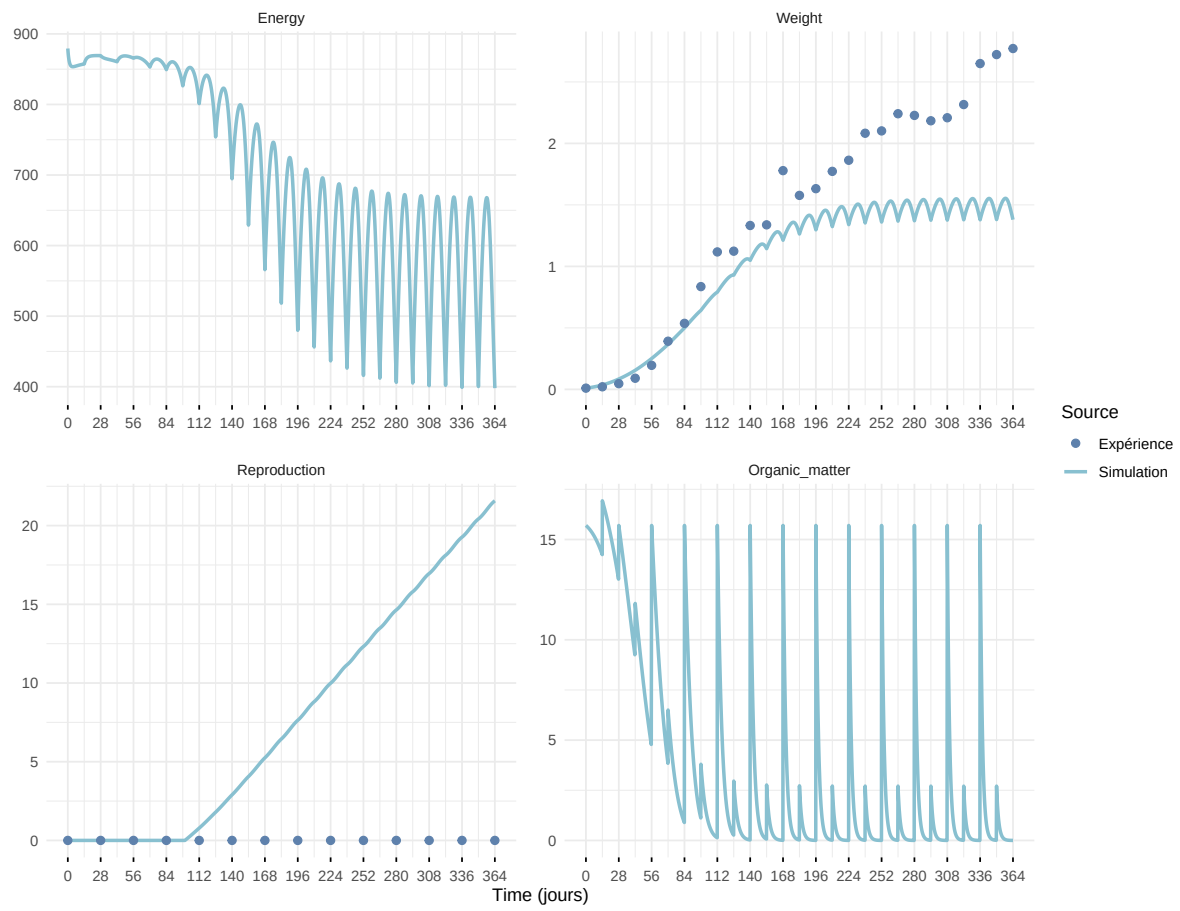
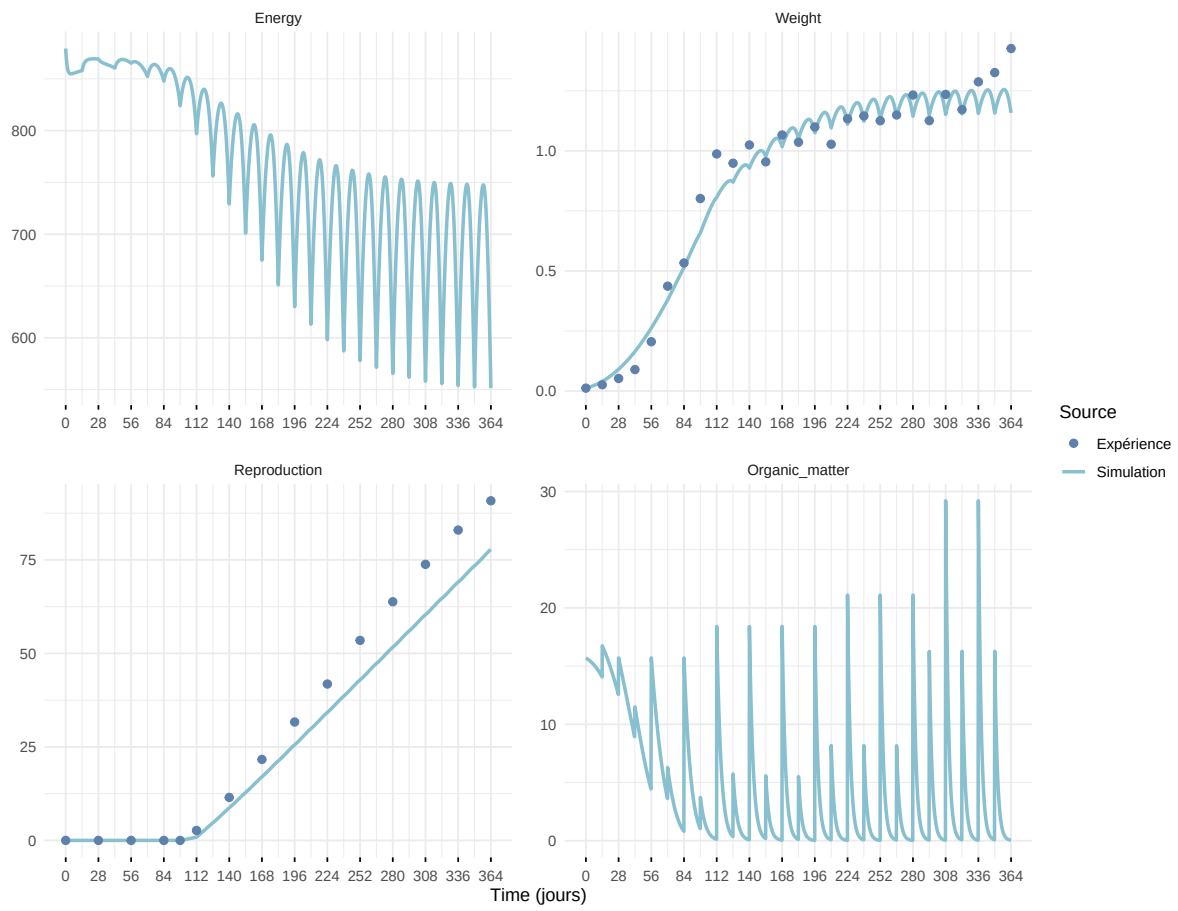


Figure 7: Simulations Weight and Reproduction.

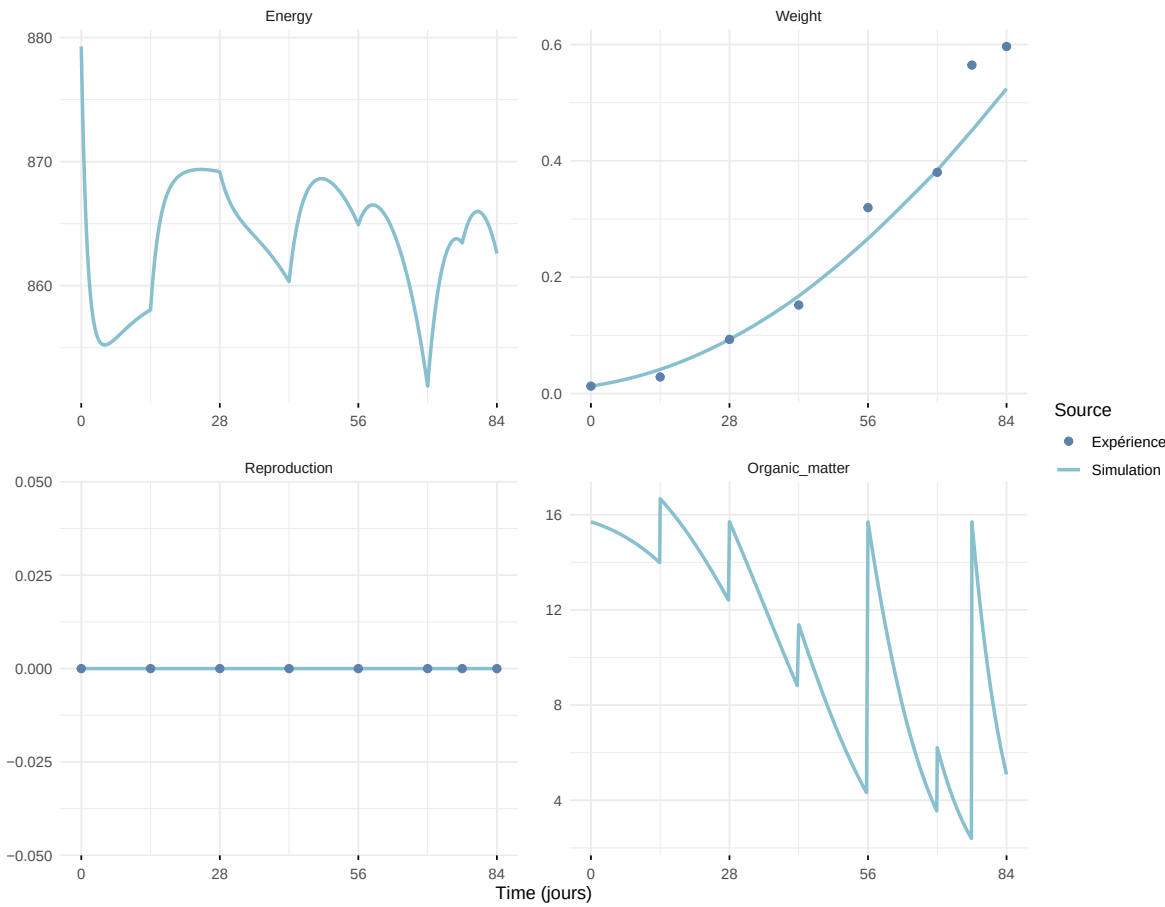
Expérience : Bart2019_XP1_1



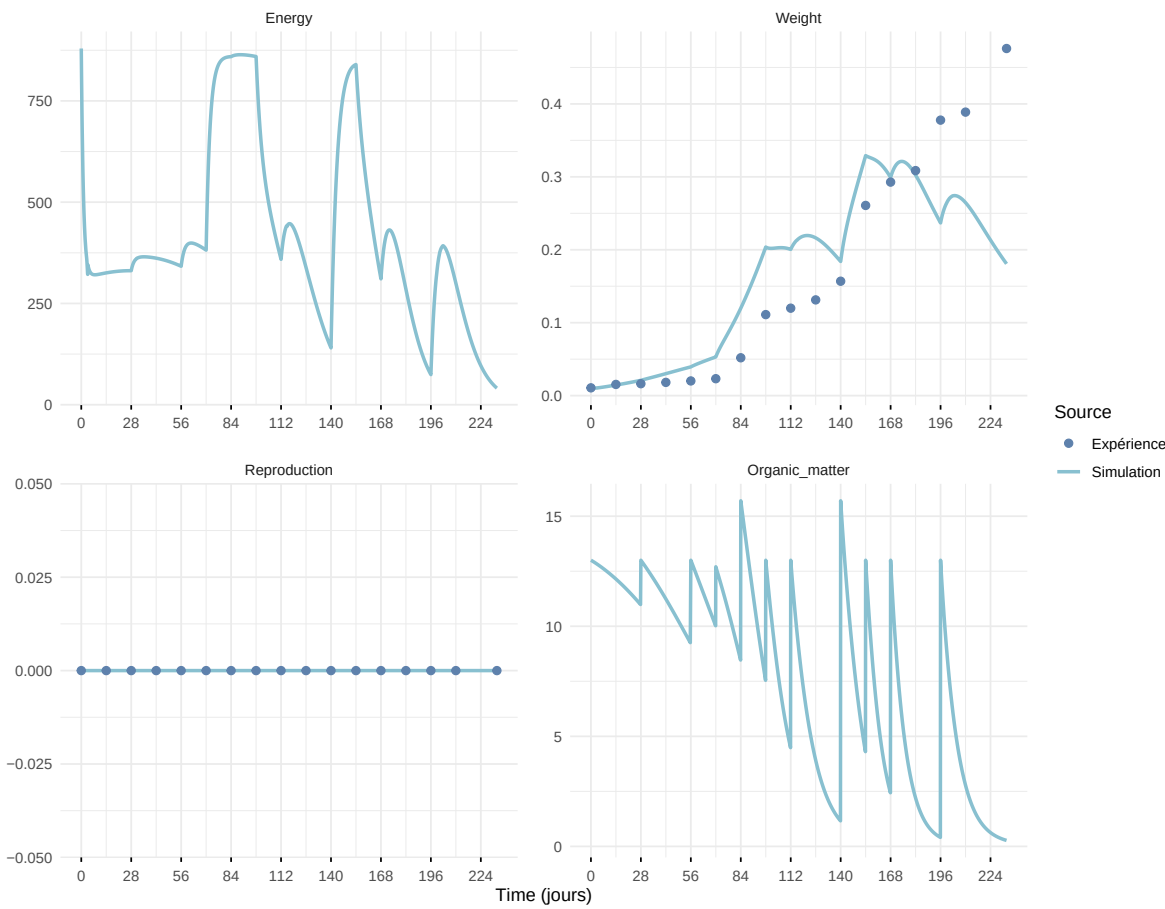
Expérience : Bart2019_XP1_2



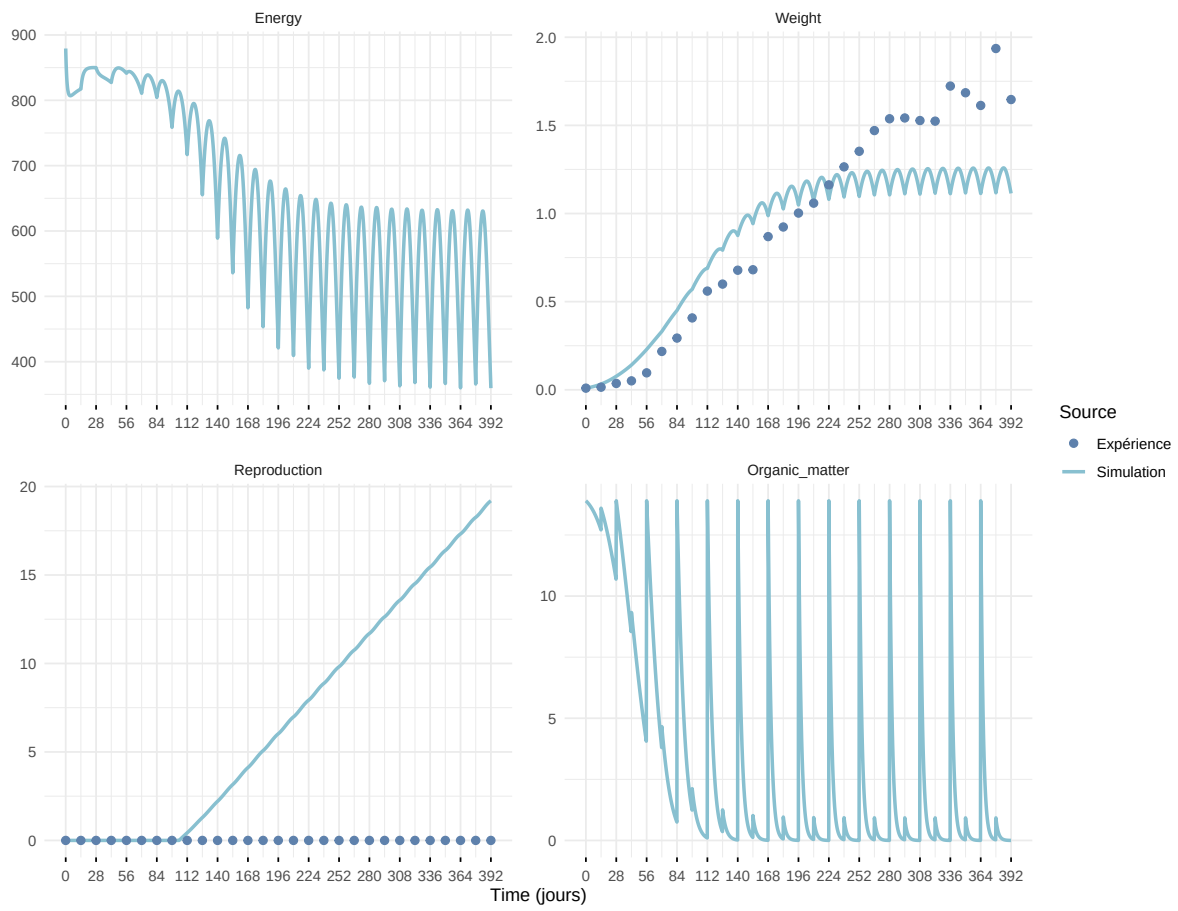
Expérience : Bart2019_XP2



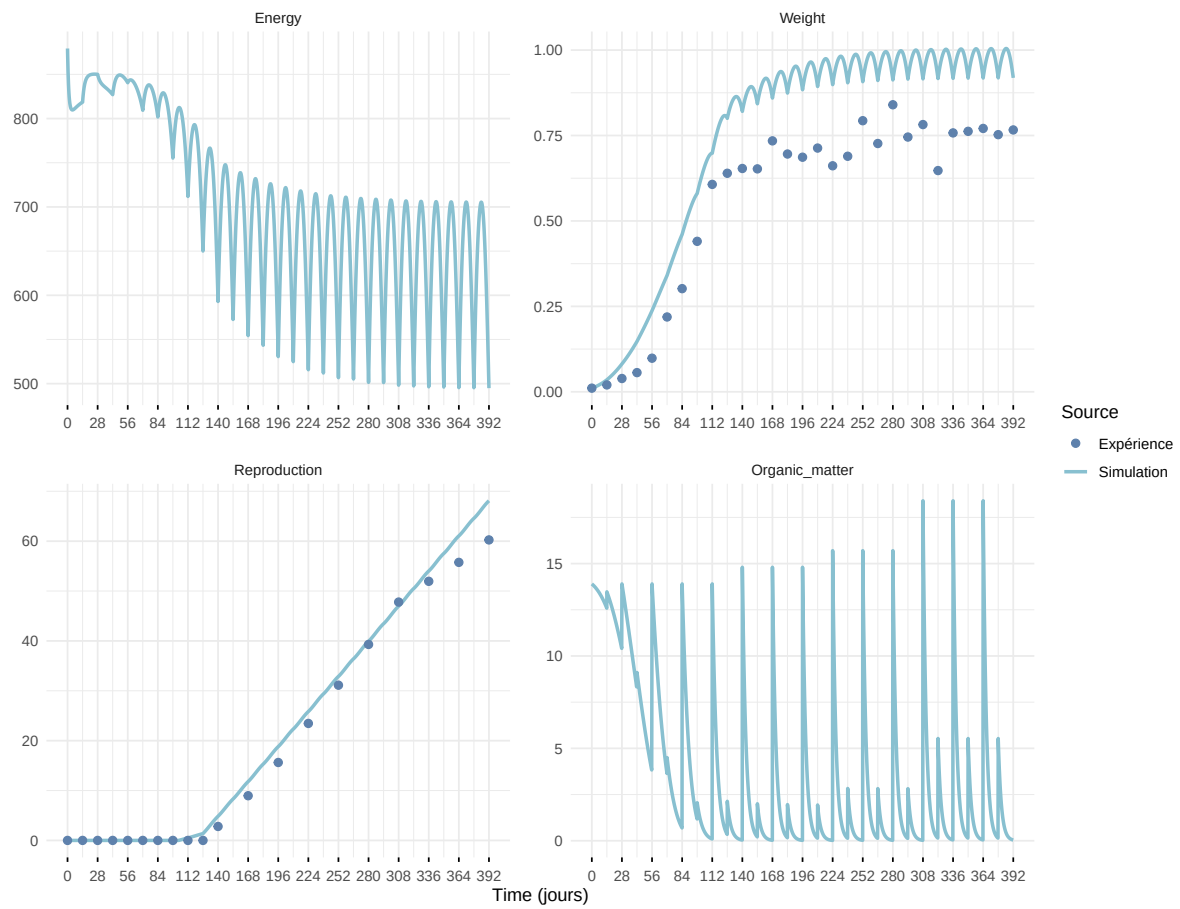
Expérience : Bart2019_XP3



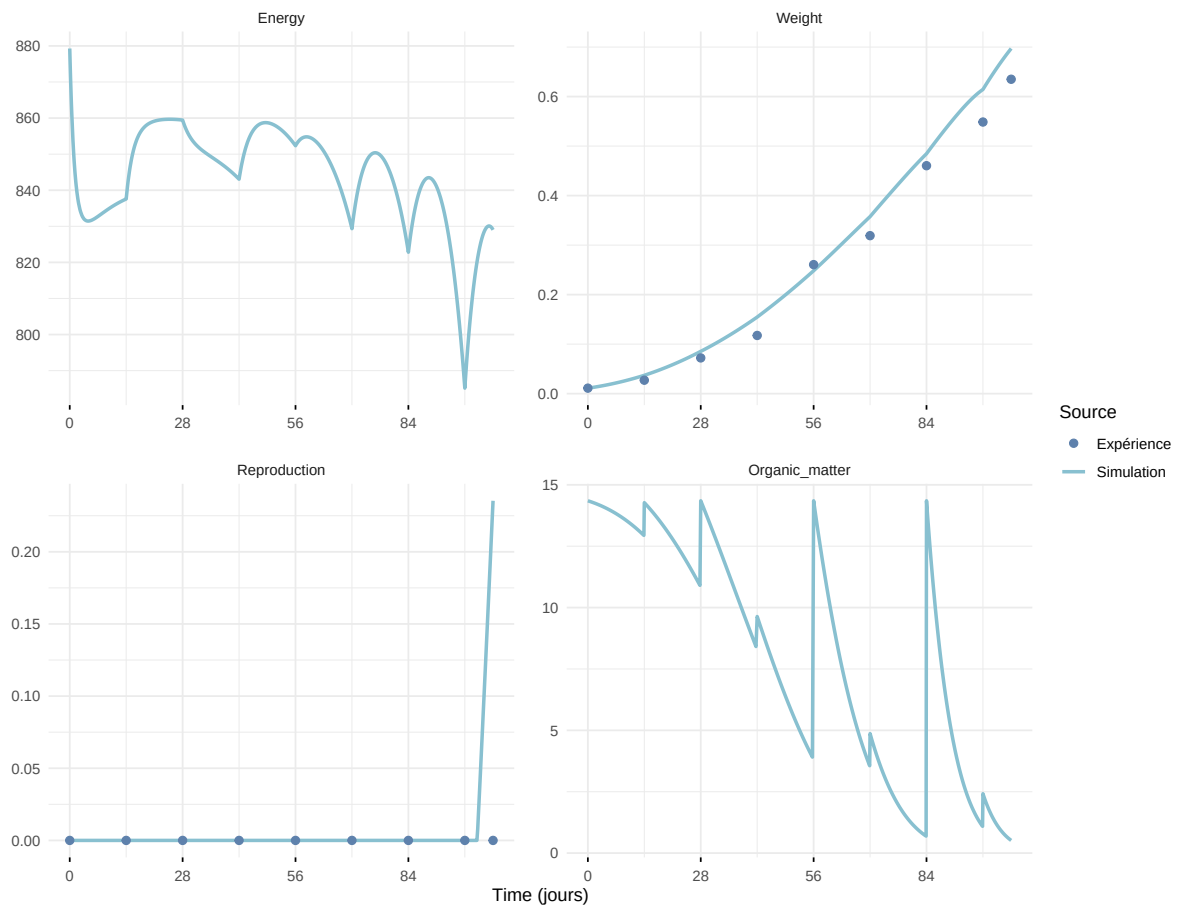
Expérience : Bart2019_XP4_1



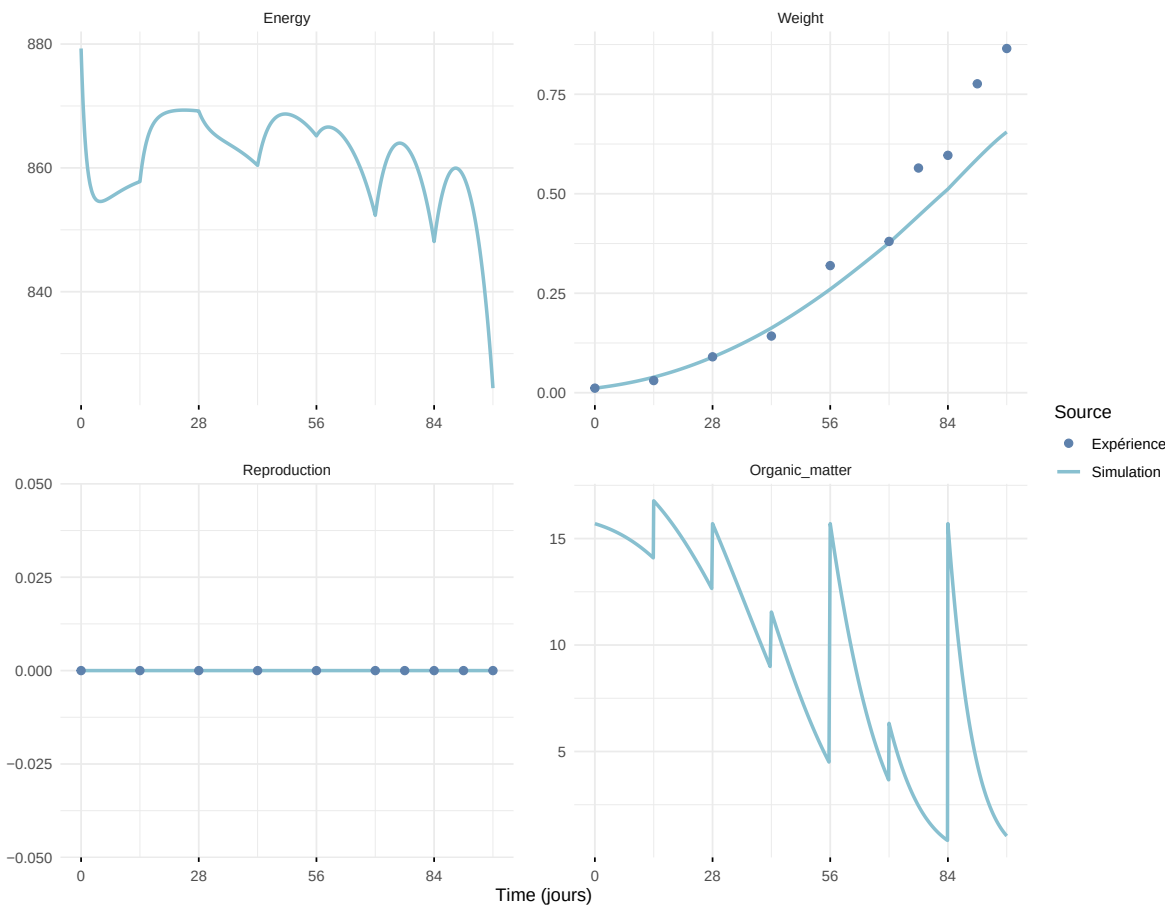
Expérience : Bart2019_XP4_2



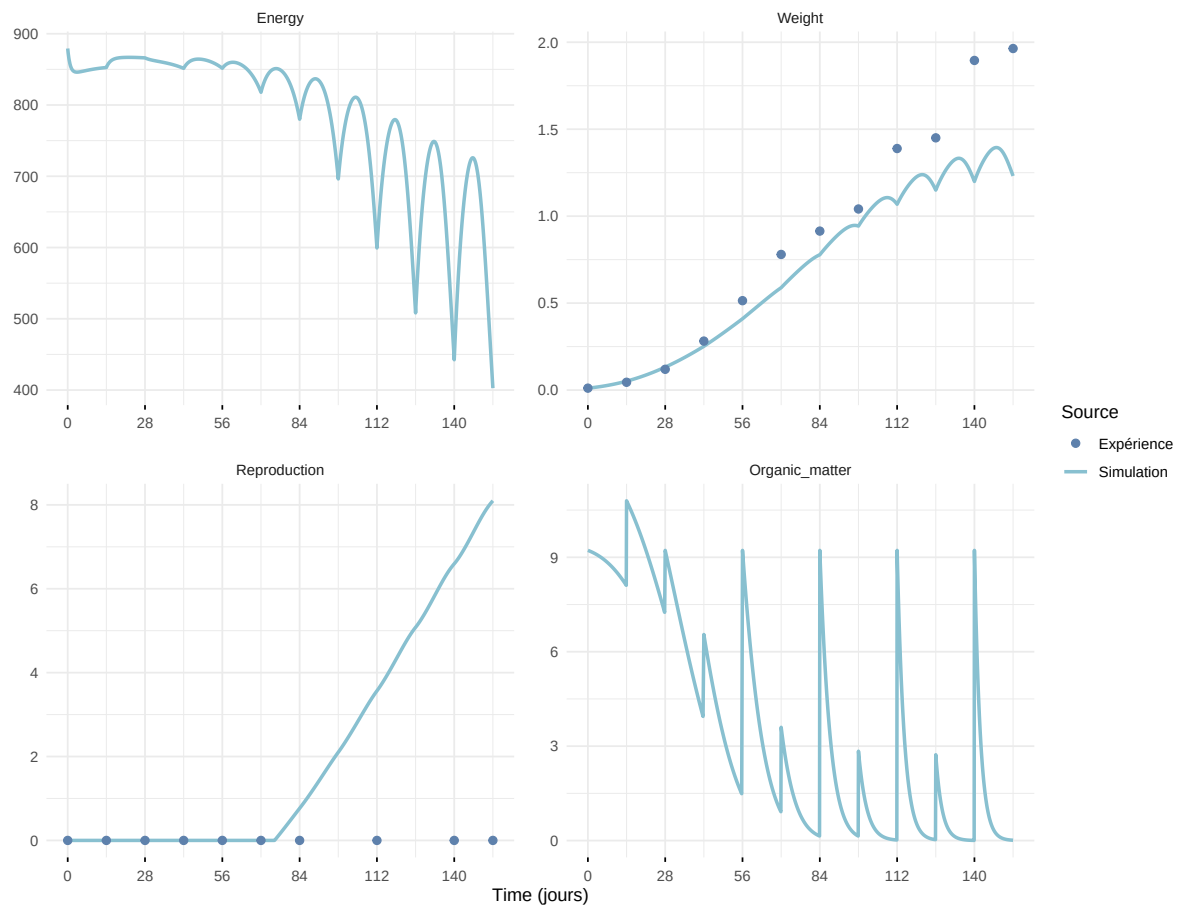
Expérience : Bart2019_XP5



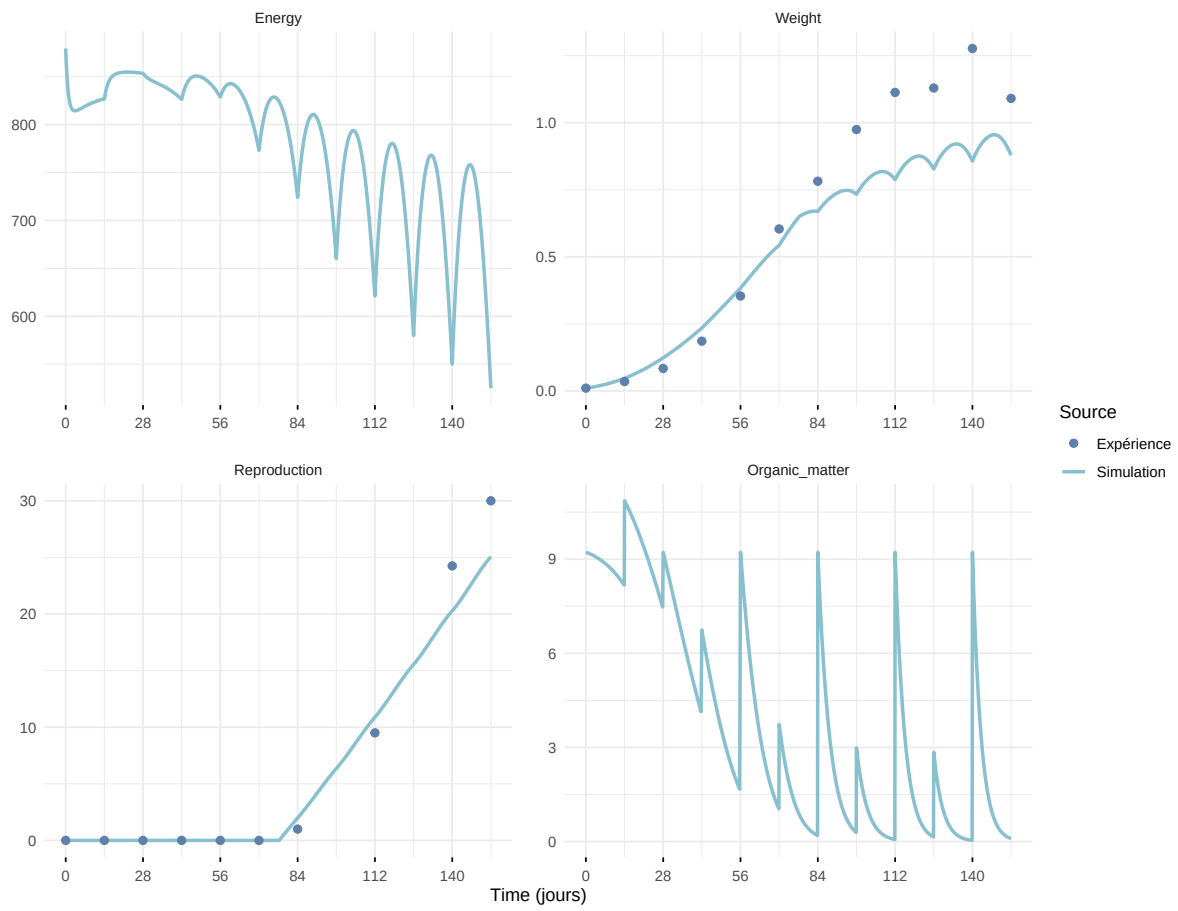
Expérience : Bart2020



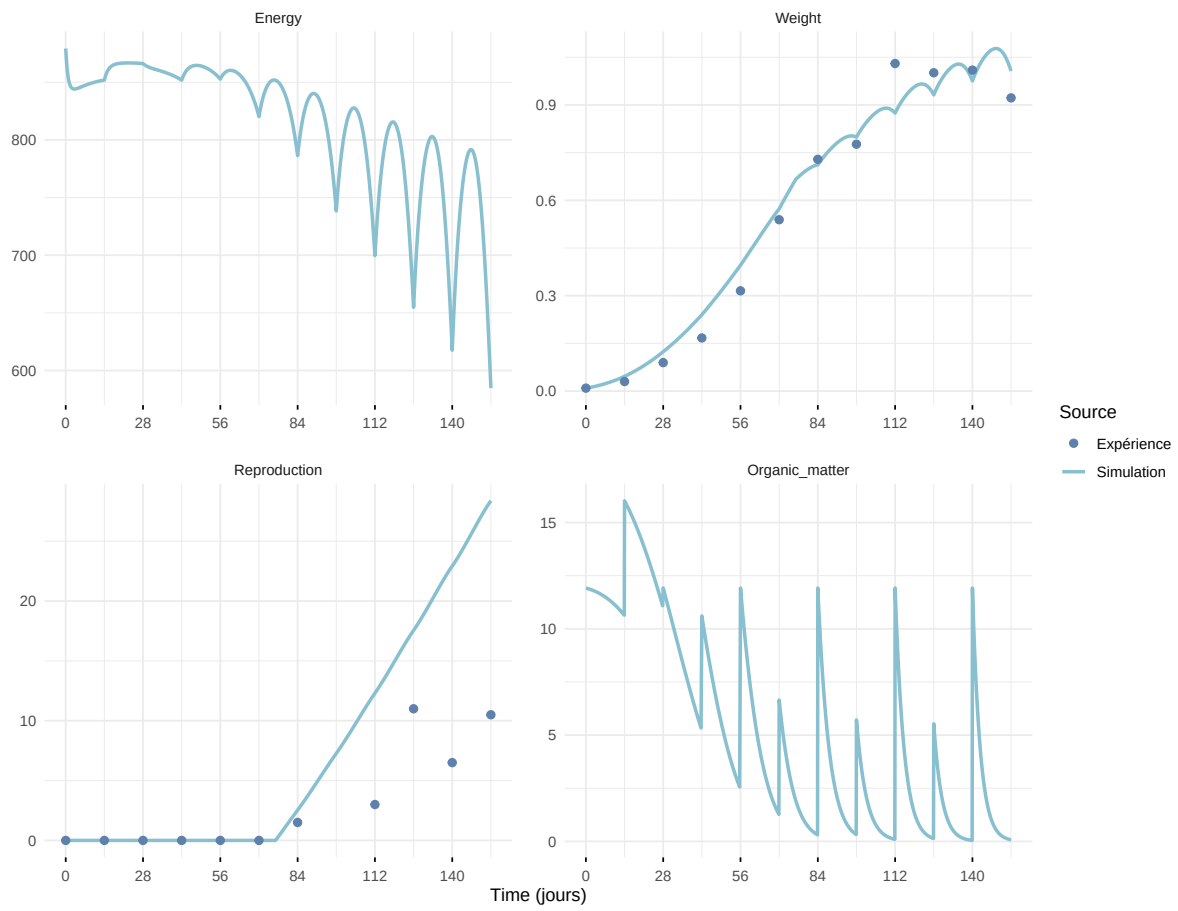
Expérience : Gollot2026_dens_D1



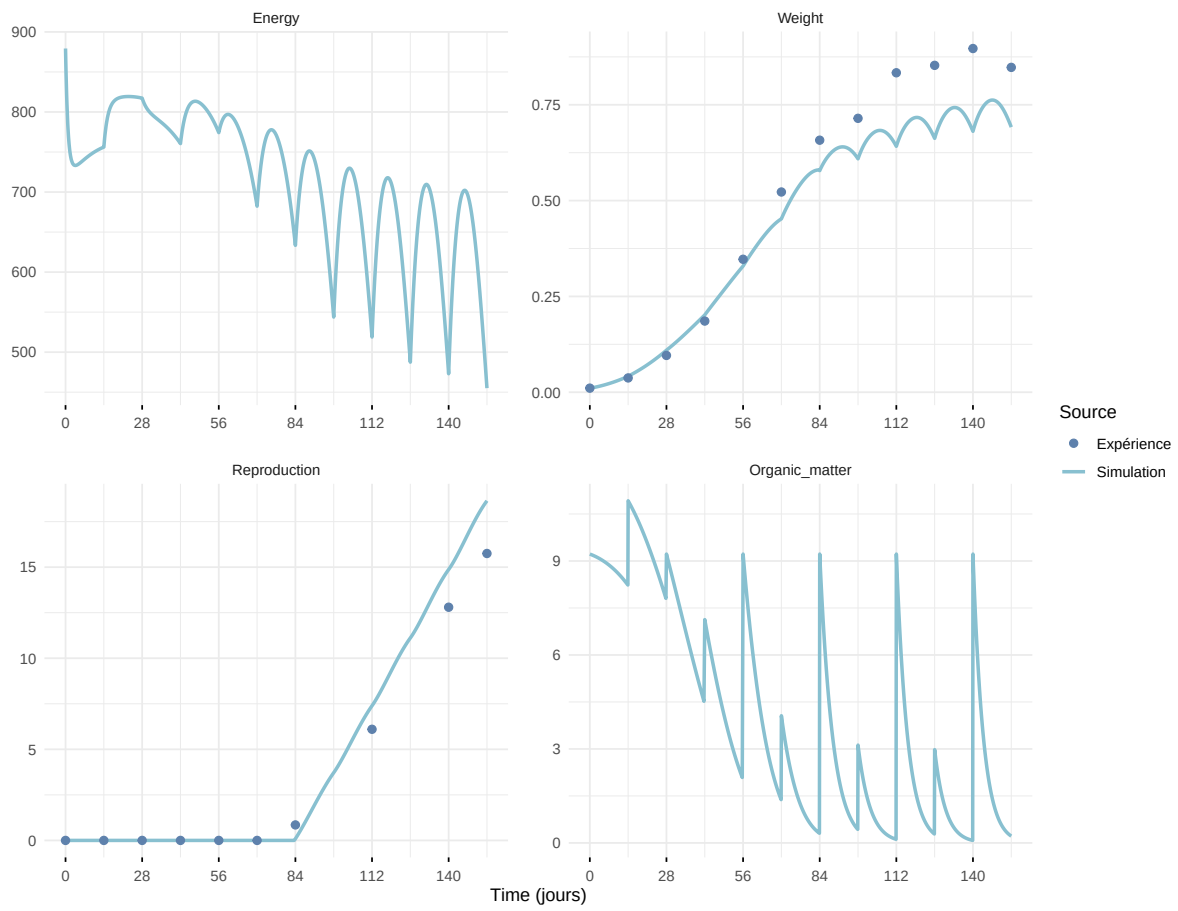
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

